

Best Time for Vacationing

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ABSTRACT

The aim of this project is to learn the Big Data and Data Mining techniques to implement them in a project chosen by our team. The name of our project is Best Time for Vacationing, and we will be using the weather dataset of the United States to analyze, compare, and show the weather trends in particular areas. Thus, we will be able to predict the weather and precipitation of those places requested by the user on a designated time range. Moreover, we will be able to suggest places that most users like. As a result, we have designed an interface that will allow users to monitor the weather of a city including its precipitation.

1. OVERVIEW

The datasets consist of the temperature, precipitation and possible humidity of some locations. These datasets have been collecting weather information for the past forty-four years. This information will help us analyze the trends of some areas over the past years, making our results more reliable. In order to make a better use of our database we have cleaned each dataset to remove the redundant and or noisy data as well as to help our system to run more efficiently. Our main goal is to make querying for the weather information easy and reliable. This can be accomplished through our software application by taking the input of users and providing them with a graphical output that can be easily interpreted. The way that our database system works is as follow: First, the user must register into our server for us to generate an id that will uniquely identify that particular user. We have included an extra feature in our project that allow users to log in through their Facebook account. This facilitates users to utilize our system in a much easier manner, and to improve our system in the registration process. Second, the user must provide us with the desired dates in which he or she wants to see temperature and or precipitation approximation for a particular city. Third, the user must provide us with the name of the city as well as the area. All cities and areas will be displayed to the user as part of

our menu. We cannot allow users to provide us with just any city but those that are in our database, we are planning to modify this feature. Third, the user contribution is very important in this last step. We will need every user to rate the cities that they have searched in order for us to create a database that will only contain the cities with the highest ranking. This way, we will be able to provide every user with a list of suggested cities that they may like according to the overall behavior of all users in our database. This last feature is also going to be combined with the actual behavior of the user and cities temperature and precipitation.

2. DESIGN CONSIDERATIONS

Best time to vacation

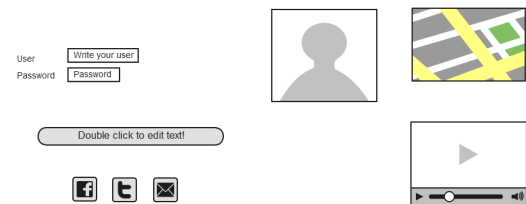


Figure 1: Design model 1.

We have designed a website that works as a platform to shape and implement our project according to our expectations. In order to fulfill the design and project patterns we are going to break down our approach in two phases that will give the reader a better insight about our approach to build our system. First. In section 2.2 we explain in details the main design of our database, which can be referenced to the tables that mainly interact with the users. In section 2.1, we enhance why the Person table was generated and explain the registration process. Also, we give a visual example, along with some query examples, of our database design built in our current implementation. Second, these sections explain better the structure of our project design. In section 3 explains how our database was obtained and how it is being used for our objective of the project. The section 4 explains how this database is implemented and developed towards the objective of our project. We also provided the

user with a visual example of how our database has been used and modeled for implementation purpose. Finally, section 5 gives an update and reflect progress on how far or close we are from reaching the goal of our project.

2.1 Registration Design

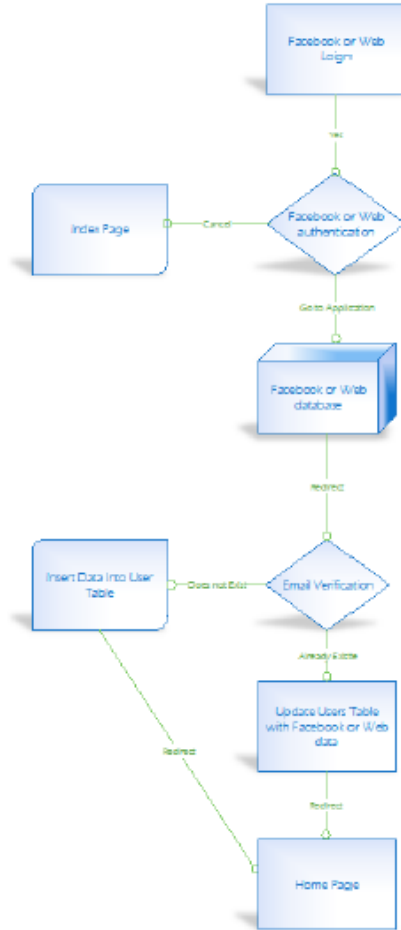


Figure 2: User Registration Flow Chart.

The table Person is ver necessary for the registration process, this can be explained as follows. First, to register the user we need the user's name and password and these data is only used for registration and user identification process. Second, we need the temperature that the user desires in a place of vacation. All these data is gathered and saved in a table named Person, and the reason why is important is because this table allows us to identify users and predict users behavior according to their search history. The other tables are better explained in section 2.2.

The registration process works in the following manner. The user needs to log in by providing a username and a password. We create a temporary table of this user to check, then we check if the user is in our system. If the user is already in our database we just update the user's table. Otherwise, we add the temporary table as a new user and update our homepage.

2.2 Database Design

We have designed a website that works as a platform to shape and implement our project according to our expectations. In order to fulfill the design and project patterns we are going to break down our approach in two sections that will give the reader a better insight about our approach to build our system. The section 3 and session 4 better explain the structure of our project design.

Person(name, password, min temperature, max temperature) Search(destination, start date, end date) Precipitation(year, month, date, longitude, latitude,min precipitation, max precipitation) Temperature(year, month, date, longitude, latitude,min temperature, max temperature) Place(City, longitude, latitude)

Tables	Attributes							
Person	name	password				minTemperature	maxTemperature	
Search	area	startDate	endDate					
Precipitation	year	month	date	longitude	latitude	minPrecipitation	maxPrecipitation	
Temperature	year	month	date	longitude	latitude	min temperature	maxTemperature	
Place	area			longitude	latitude			

Figure 3: Main Entity Tables.

```

CREATE TABLE "searchPlace_place" (
  "id" serial NOT NULL PRIMARY KEY,
  "city" varchar(2) NOT NULL,
  "slug" varchar(50) NOT NULL UNIQUE,
  "longitude" numeric(5, 2) NOT NULL,
  "latitude" numeric(5, 2) NOT NULL
);

```

Figure 4: User Search Place Table.

Figure 4 shows how our search place table looks like. We have assigned to each area, which we named city in our query, a unique id through which we can distinguish it from the other areas. Each city has its own longitude and latitude and these values are provided by our database. The slug is required by our system, this is a way of generating a valid URL using the data already obtained.

```

--
-- Table structure for table `city`
--
DROP TABLE IF EXISTS `city`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `city` (
  `name` varchar(25) DEFAULT NULL,
  `latitude` float DEFAULT NULL,
  `longitude` float DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
/*!40101 SET character_set_client = @saved_cs_client */;

```

Figure 5: Place Table.

Figure 5, this table is not as important as the others. However, it has a small factor of influence when predicting either temperature or precipitation of an area. The attribute

values of these tables are the longitude and latitude of an area, and these data is obtained from our database.

```
CREATE TABLE "searchPlace_customfacebookuser" (
  "id" serial NOT NULL PRIMARY KEY,
  "password" varchar(128) NOT NULL,
  "last_login" timestamp with time zone NOT NULL,
  "is_superuser" boolean NOT NULL,
  "username" varchar(30) NOT NULL UNIQUE,
  "first_name" varchar(30) NOT NULL,
  "last_name" varchar(30) NOT NULL,
  "email" varchar(75) NOT NULL,
  "is_staff" boolean NOT NULL,
  "is_active" boolean NOT NULL,
  "date_joined" timestamp with time zone NOT NULL,
  "about_me" text,
  "facebook_id" bigint UNIQUE,
  "access_token" text,
  "facebook_name" varchar(255),
  "facebook_profile_url" text,
  "website_url" text,
  "blog_url" text,
  "date_of_birth" date,
  "gender" varchar(1),
  "raw_data" text,
  "facebook_open_graph" boolean,
  "new_token_required" boolean NOT NULL,
  "image" varchar(255),
  "state" varchar(255)
);
```

Figure 6: Facebook User Search Place Table.

Figure 7 is another table that we are currently using in our implementation. This table is only users with Facebook users that log into our system. The more important attributes in this table are the id, password, username and email. These are the attributes that we are using the most in our queries to relate every search and or ranking of a city to the user.

```
--
-- Table structure for table `precipitation`
--
DROP TABLE IF EXISTS `precipitation`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `precipitation` (
  `year` int(11) DEFAULT NULL,
  `month` int(11) DEFAULT NULL,
  `day` int(11) DEFAULT NULL,
  `longitude` float DEFAULT NULL,
  `latitude` float DEFAULT NULL,
  `precipitation` float DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
/*!40101 SET character_set_client = @saved_cs_client */;
```

Figure 7: Precipitation Table.

Figure 8 and 9 are the tables representation of those used in our implementation. Temperature and precipitation tables are very similar, except for the fact that in temperature we deal with max and min temperatures and in precipitation we deal with max and min precipitations. These are two of the most important tables in our database, and it is through them that we are going to predict the temperature and or precipitation of an area searched users. Also, these

```
--
-- Table structure for table `temperature`
--
DROP TABLE IF EXISTS `temperature`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!40101 SET character_set_client = utf8 */;
CREATE TABLE `precipitation` (
  `year` int(11) DEFAULT NULL,
  `month` int(11) DEFAULT NULL,
  `day` int(11) DEFAULT NULL,
  `longitude` float DEFAULT NULL,
  `latitude` float DEFAULT NULL,
  `temperature` float DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
/*!40101 SET character_set_client = @saved_cs_client */;
```

Figure 8: Temperature Table.

tables could be very helpful to help us to get a more accurate weather prediction.

3. ARCHITECTURE

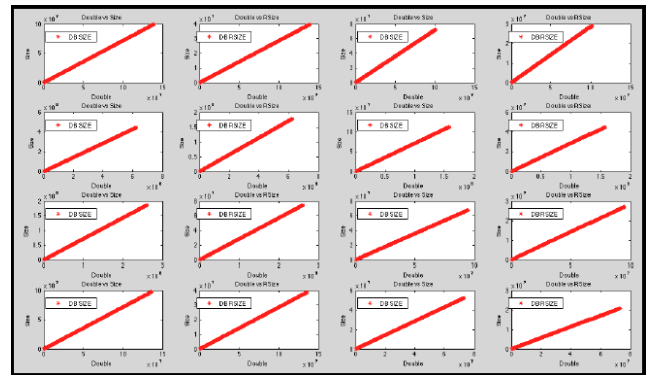


Figure 9: Cleaning Process and CPU efficiency

PostgreSQL is the database engine underlined by our management system to create, read, update and delete data from a database. Moreover, we have included other database and web development software that have helped us to model and process our database faster and easier. The application-programming interface that we are using, to allow users to interact with our underlying engine without going through the user interface of our system, is Django. We are also making use of tools such as Python and HTML. On the other hand, we have used data cleaning-techniques, not related to the above tools, to clean all datasets downloaded from the Internet and convert them into a useable and more meaningful database.

The cleaning process can better be explained through the figure 9, this shows the difference of the CPU processing approximation time between the data before cleaning and the data after cleaning. Unfortunately, because of the space and time we only provided one image of the space and time efficiency gained. All Size graphs are the sizes of the data of the areas, or cities, in our database before cleaning, and

all RSize graphs are the data of the the same areas after the cleaning process. If you see carefully you can see that the RSize database is eighteen times faster than the Size graphs. This is only for figure 9, the overall space and efficiency gained is about 3

3.1 Work Platform

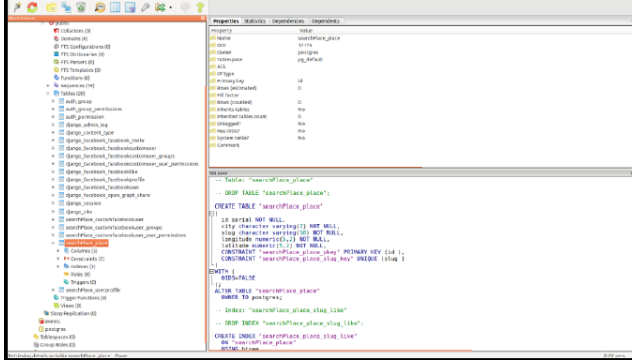


Figure 10: pgAdminIII

We have obtained weather and precipitation information of some areas in the United States for our database. We were able to obtain temperature information, from most of the states, that has been gathered since 1980 until 2000. We must emphasize that collecting datasets from the United States has been a time consuming task, and this has precluded us from gathering weather datasets from 2000 until 2012. However, we are planning to collect more recent weather information, from 2000 till 2012, to include it in our database once our prototype is working as desired.

The platform that we are using to manage our database is Django. The installation and upgrading process is very easy since we have the ability to automatically install database adapters and other dependencies. Using Django in our management systems has allowed us to extract valued content from our data sets and build a high performing web applications system in a elegant and efficient way. The following factors have been improved with the use of Django. First, we can define our data models much easier with Python that provides us with a dynamic database access APIA, which we have been writing with SQL. Second, it provides us with an automatic admin interface that adds and updates users and user's content automatically. Third, it supports multi-language applications. This is essential for our implementation since is mostly for vacation purpose. Finally, we can always use cache frameworks to increase future performance.

4. IMPLEMENTATION

The implementation of our project has already taken place since most of the weather and precipitation dataset from most cities of the United States was gathered. This data is used for analysis purpose to give the user a weather approximation, including precipitation and possibly humidity, on a range of dates designated by the user. Every user must log in, or sign up, to have a unique id and password. This feature has already been built and can be seen in figure 12. This is an application for users to log into our system through Facebook. This feature facilitates users to utilize our system

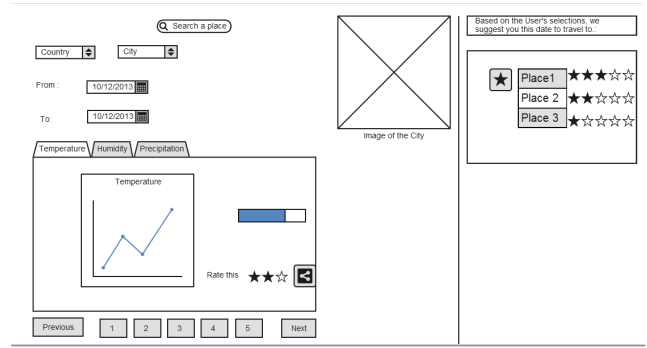


Figure 11: Implementation design model 2.

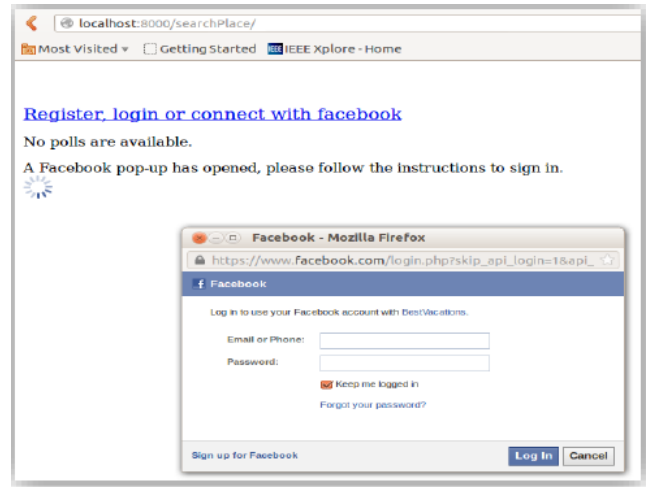


Figure 12: Implementation design model 2.

in a much easier manner as well as to improved our system as follows. First, helping us to increase our project registration. Second, providing us with app id and app secret id, this way the user does not have to worry about creating a new username and or password. Overall, this feature could potentially help us to integrate more features and for future improvements. The dataset for the cities along with the user dataset is being used to create another dataset that only contains with the highest ranking given by the users. Figure 13 is another feature that we have already built is main site. This is the view that users get after the log into our system. Once they log in, they can either like a city or add a city. Also, users can open the graph to see a graphical representation of the approximated weather and precipitation of the designated of the places they haver searched. Figure 14 is another feature that we have built as well, this is how a user can search for a city of his or her interest.

5. CURRENT STATUS & FUTURE WORK

We have gathered weather and precipitation information of the United States for our database. We were able to obtain temperature information, from most of the states, which has been collected from 1980 until 2000. We most emphasize that the task of collecting datasets from the United

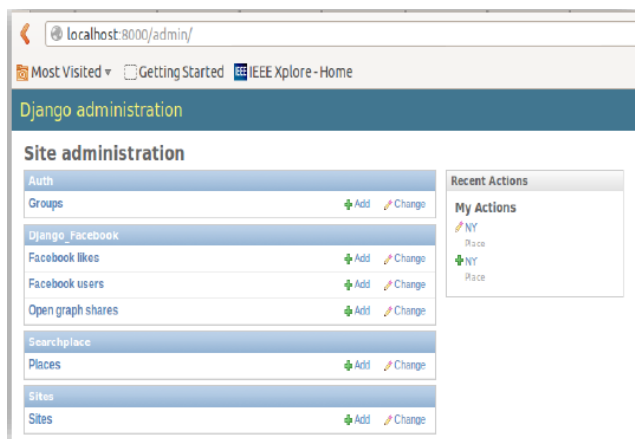


Figure 13: Implementation design model 2.

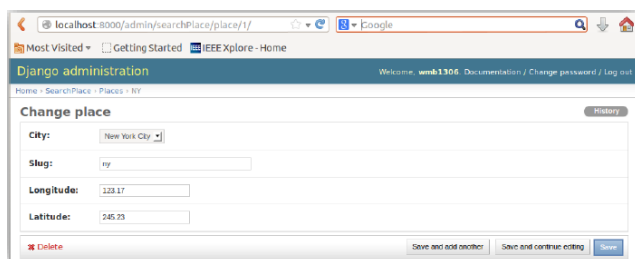


Figure 14: Implementation design model 2.

States has been time consuming, and it has precluded us from gathering weather datasets from 2000 until 2012. However, we will proceed to collect more recent weather information from the time period mentioned above and include it in our database once our prototype is working as desired.

Recently, we have developed an interface as the platform where all this data manipulation and managing is taking place. We are currently able to take a user input, such as logging information. Next, all sample datasets that have been downloaded has been cleaned to remove the unwanted information, which has been converted into usable SQL database. In addition to this, we are currently asking the user to designate a city, and working on systematically approximate the future temperature and or precipitation of this. Finally, we started developing an extra feature for our application. This feature is based on users previous selected locations and their ranking to prioritize the list of suggested places that users may like based on our system users overall behavior. The suggested places are chosen comparing the ranking of locations that were previously selected and analyzing the overall system and actual user behavior.

On the other hand, we have not yet implemented some desired features for our project. These features are weighted average and humidity,. First, the weighted average would use a more efficient technique to approximate a temperature and precipitation even more accurately through a comparison system that will be able to prioritize years according by testing if the either have a balanced temperature behavior and or are closest to the recent year. Second, we are working

on gathering a dataset with humidity information regarding to the locations or places that we have gathered so far, which would give the user a better idea of how the weather really feels in certain location. Once we have achieved the criteria mentioned above, we will proceed to expand our database by adding the datasets of other countries. These countries include Ecuador, the Dominican Republic, and India, whose datasets will be used to implement the above strategy to provide users with a diversity of locations that best suit their requirements.

5.1 Important point that we have addressed

1) design of the data base

Person(name, password, min temperature, max temperature). Search(destination, start date, end date). Precipitation(year, month, date, longitude, latitude,min precipitation, max precipitation). Temperature(year, month, date, longitude, latitude,min temperature, max temperature). Place(City, longitude, latitude).

2) type of queries

Ranking a city - outside. searching for a city - outside. Selecting the city with the highest ranking - inside. Selecting the city viewed, or chosen, by the user the most. - inside. searching for the temperatures of that area within the last 30 years - inside. Creating a list, table, with the cities ordered according to their ranking - inside. Calculating average temperatures to predict a close approximation to a area's temperature - inside. Calculating average precipitations to predict a close approximation to a area's precipitation - inside.

3) explain reflect progress User registration . searching for a city. ranking a city. selecting viewed cities (not the most, not yet).

4) changes for the final submission We are still not sure, but if we run out of time we might not have our web design to look exactly as our model. However, we are curtailed that it will look nice. Also, if we do not find enough data to add for humidity prediction we might exclude humidity out and only work with temperature and precipitation.

5) type of data analyses to be performed Predicting a city temperature. Predicting a city precipitation . Generating a personalized list of suggested places for users. Generate a list of most ranked places.

5.2 Tables, Figures, and Citations/References

The tables below contain the datasets that we have obtained. However, they only represent a summary of the temperatures from some of the stations in the United States. It is important to highlight that each one of the station, or even a summary of all stations, contain huge data. Due to the large size of the data obtained by stations that are gathering weather information, it would be impossible for us to attach their respective datasets in the project. Therefore, we only attached the data processing scale in figure 9. If you look close, this can show you not just how much our database has improved but how big it is as well.

6. CITATIONS/REFERENCES

<http://www.ncdc.noaa.gov/cdo-web/>
<http://www.ncdc.noaa.gov/cdo-web/datasets>